

Varshaa Sekar Jayanthi

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Software Engineer | 2+ years full-time in production C#/.NET and Python

EDUCATION

Northwestern University, McCormick School of Engineering

Chicago, IL

Master of Science in Information Technology | GPA: 3.75

Sep 2025 – Dec 2026

- Computer Science coursework: Machine Learning, Deep Learning, Scalable Software Architecture, Algorithms

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Information Systems | Minor in Business Administration

Aug 2019 – May 2023

- Computer Science coursework: 15-112 Fundamentals of Programming, 15-121 Data Structures (Java), 17-313 Software Engineering
- Awards: Grace Hopper Celebration Scholarship, Senior Leadership Recognition Award

EXPERIENCE

Tenacitics, Inc.

Houston, TX (Remote)

Software Engineer Intern | Internal SQL deployment platform

Jun 2026 – Present

- Built the platform in Python, replacing a 3-system ticket-and-email handoff and cutting release effort **68%** across **120** developers
- Eliminated partial writes across all **4 environments**: transactional execution, enforced timeouts, API-level role separation
- Proved that a failed migration leaves the database unchanged: **113 test cases**, **98%** coverage against a live PostgreSQL 16 instance

Northwestern University

Chicago, IL

Graduate Researcher | Agentic systems evaluation | Python, Kubernetes, Docker, LLM

Jan 2026 – Jun 2026

- Ran the ablation isolating verifier-grounded memory across **400+** GPT-5.5 evaluations, lifting accuracy **54% to 78.5%**
- Identified environment setup as the dominant failure cause, then co-built the **10-attempt** loop that lets agents recover

Computer Enterprises Inc.

Pittsburgh, PA

Software Developer, embedded vendor-side owner | Client: Zelis Payments, large-scale healthcare platform

Jun 2024 – Aug 2025

- Shipped **10 major features** across claims pricing and payment workflows in C#/.NET and React
- Improved database performance **40%** by rewriting slow queries, restructuring joins, adding indexes, and narrowing SELECT * scans
- Built an LLM pipeline that automated verification across **10+** enterprise data migrations at **98%** accuracy

UPMC Corporate Services

Pittsburgh, PA

Software Engineer | C#/.NET, ASP.NET MVC, MS SQL Server, Azure DevOps

Jul 2023 – Jun 2024

- Developed **~20 features** on a **\$1.5M** Health Plan web application and presented it to the CTO, spanning new API endpoints, service integrations, stored procedures, schema changes and Agile delivery
- Lowered production error rates **47%** with regression suites gating releases; owned the lower-environment database

Carnegie Mellon University

Pittsburgh, PA

Teaching Assistant | Application Design and Development

Jan 2023 – Aug 2023

- Taught weekly labs to **80+** students in a required core CMU course: MVC, relational database design, and full-stack Rails

PROJECTS & COMPETITIONS

Prediction Market Order Book & Market-Making Simulator | Python; C++17 port in progress

May 2026 – Present

- Built an event-driven limit order book over live prediction markets, with $O(\log n)$ insertion and $O(1)$ cancellation
- Measured what adverse selection costs a market maker: break-even at a **4.7c** half-spread over **20 seeds** per point

IMC Prosperity 4 Global Algorithmic Trading Competition | Python

Apr 2026 – May 2026

- Team captain; reached the final round, ranking **1,275 of 18,803 teams (top 7%)**. Built the market-making engine and **doubled** per-day profit on the highest-volume product by replacing a static fair value with a rolling-mean estimator
- Priced the options round from scratch with **Black-Scholes** and **Newton-Raphson implied volatility** solved on vega, quoting each strike off the anchor contract's live IV; also traded sum-constraint basket arbitrage across a linked product group

BluePrint | Early-stage startup, AI-orchestrated SDLC governance platform; TypeScript, PostgreSQL, Node.js

Jul 2026 – Present

- Accelerated a knowledge-graph subgraph query from **31s to 183ms** by diagnosing fan-in convergence in a recursive CTE, cutting intermediate rows from **472,421 to 123**; output verified set-identical
- Closed **2 live** cross-tenant data-exposure paths, one a state-changing write, by root-causing parameter loss in Express middleware; audited **22 routers** for blast radius

LEADERSHIP

Carnegie Mellon Student Government, both positions elected, 2021 to 2023:

Vice President, Activities Board: grew membership **80 to 200**, owning budget and programming.

Lead Financial Advisor, Joint Funding Committee: allocated **\$2.3M** across **300+** organizations against \$4.2M requested.

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript/JavaScript, SQL, Java, Ruby, Swift, HTML/CSS

Frameworks & APIs: .NET, ASP.NET, React, Node.js, Express, FastAPI, Rails, REST APIs, OOP, Agile/Scrum

Infrastructure & DevOps: AWS (Lambda, API Gateway), Azure, DevOps, CI/CD, Docker, Kubernetes, Linux, Git/GitHub

Data & ML: PostgreSQL, MS SQL Server, MongoDB, LLM evaluation, regression, data and statistical analysis, NumPy, Pandas, scikit-learn